

MedConnect Hub - Product Specification

1. Purpose

MedConnect Hub is a healthcare coordination platform designed to connect patients, doctors, hospitals, labs, and administrators in one secure digital workflow.

Its goal is to reduce delays, improve care continuity, and make medical operations transparent for all stakeholders.

2. Product Vision

Create a single reliable system where:

- Patients can access care quickly and track their health journey.
- Providers can manage consultations, records, and follow-ups efficiently.
- Hospitals can coordinate departments, resources, and outcomes in real time.

3. Target Users

- Patients
- Doctors
- Nurses and clinical staff
- Hospital administrators
- Lab technicians
- Pharmacists
- Support and compliance teams

4. Core Features

4.1 User and Access Management

- Secure sign-up/login with role-based access.
- User profiles for patients and staff.
- Permission controls by role (patient, doctor, admin, lab, pharmacy).
- Password reset and account recovery.

4.2 Patient Management

- Patient onboarding with demographic and contact information.
- Digital medical profile (history, allergies, chronic conditions).
- Visit timeline with previous consultations and outcomes.
- Emergency contact and insurance details.

4.3 Appointment and Scheduling

- Book, reschedule, and cancel appointments.
- Doctor availability calendar with slot management.
- Queue visibility for same-day appointments.
- Automated reminders (SMS/email/in-app).

4.4 Consultation Workflow

- Consultation notes entry and updates.
- Diagnosis and treatment plan capture.
- Prescription generation and history tracking.
- Follow-up recommendation and scheduling.

4.5 Electronic Medical Records (EMR)

- Centralized patient record storage.
- Upload and attach medical files (reports, scans, prescriptions).
- Structured entries for diagnosis, vitals, and medications.
- Audit trail for all record changes.

4.6 Lab and Diagnostics Integration

- Lab test ordering from consultation.
- Lab dashboard for pending and completed tests.
- Result upload and doctor notification.
- Patient-visible lab result history.

4.7 Pharmacy and Medication

- E-prescription workflow from doctor to pharmacy.
- Medication fulfillment status tracking.
- Refill request process for eligible prescriptions.
- Drug interaction warning support (if enabled).

4.8 Billing and Payments

- Service invoice generation (consultation, lab, pharmacy).
- Payment status tracking (paid, pending, failed).
- Insurance claim preparation support.
- Basic revenue and transaction summaries.

4.9 Notifications and Communication

- Appointment confirmations and reminders.
- Alerts for lab results, prescriptions, and follow-ups.
- In-app messaging between patient and care provider (optional scope).
- Escalation alerts for urgent care events.

4.10 Reporting and Analytics

- Operational dashboards (appointments, waiting time, utilization).
- Clinical metrics (follow-up rates, common conditions).
- Financial summaries (daily/monthly billing).
- Exportable reports for management.

5. How the Product Should Work (End-to-End)

5.1 Patient Journey

1. Patient registers and completes profile.

2. Patient books an appointment with a suitable doctor.
3. System sends confirmation and reminder notifications.
4. Doctor conducts consultation and records notes/diagnosis.
5. Doctor orders lab tests and/or prescriptions when needed.
6. Lab uploads results; patient and doctor are notified.
7. Pharmacy fulfills prescription and updates status.
8. Billing is generated and payment is tracked.
9. Follow-up appointment is scheduled and monitored.

5.2 Provider Journey

1. Doctor logs in and reviews daily appointment list.
2. Doctor opens patient chart and history before consultation.
3. During consultation, doctor enters findings and plan.
4. Doctor places test orders and prescriptions digitally.
5. Doctor reviews incoming lab results and adjusts treatment.
6. Doctor closes visit with follow-up instructions.

5.3 Admin Journey

1. Admin manages users, roles, and access permissions.
2. Admin monitors appointments, queues, and department load.
3. Admin tracks billing, payments, and operational KPIs.
4. Admin exports reports for compliance and decision-making.

6. Functional Requirements

- The system must support multi-role authentication and authorization.
- The system must maintain complete patient history per profile.
- The system must allow appointment scheduling with conflict prevention.
- The system must support consultation notes, test orders, and prescriptions.
- The system must store lab results and attach them to patient records.
- The system must generate billing records tied to medical services.
- The system must log user actions for audit and compliance.

7. Non-Functional Requirements

- Security: encryption in transit and at rest for sensitive data.
- Privacy: role-limited access to medical information.
- Reliability: high availability during hospital working hours.
- Performance: common user actions should respond quickly.
- Scalability: support growth in patients, providers, and records.
- Usability: simple workflows for both technical and non-technical users.

8. Success Criteria

- Reduced appointment no-show and scheduling delays.
- Faster consultation-to-treatment cycle.
- Improved record completeness and fewer documentation gaps.
- Better visibility into operational and clinical performance.

- High user adoption across patient and provider roles.

9. Suggested Future Enhancements

- Telemedicine/video consultations.
- AI-assisted triage and clinical decision support.
- Wearable and remote monitoring integration.
- Multi-hospital network support.
- Mobile-first patient and provider apps.